Leander F. Thiele

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EMPLOYMENT

Kavli IPMU, University of Tokyo 2024–present
Project Assistant Professor (tenure-track) in Center for Data-Driven Discovery (CD3)

EDUCATION

Princeton University, Physics, PhD 2019–2024
advisor: David N. Spergel
thesis: *Getting ready for new Data: Approaches to some Challenges in Cosmology*

Perimeter Institute, Theoretical Physics, MSc 2018–2019
advisor: Kendrick M. Smith, co-advisor: J. Colin Hill
thesis: *Capturing non-Gaussianity: Analytic model for the one-point probability distribution function of cosmological fields within the halo model*

University of Oxford, Physics, BA First Class 2015–2018
ranked top of the cohort ($\sim$ 130 students) in years 2 and 3

HONORS

Kusaka Memorial Prize in Physics (Princeton)	$3000	2022
Member of the German Academic Scholarship Foundation	€35k	2015–2019
Perimeter Scholars International Award (Perimeter)	CA$45k	2018
Scott Prize for best performance in the 3rd year (Oxford)	£400	2018
Winton Capital Prize for best performance in the 2nd year (Oxford)	£250	2017
BP Scholarship (Oxford)	£2000	2017
Rokos Award for summer research project (Oxford)	£800	2016

FUNDING

JST PRESTO JPMJPR26287514, *Human-AI Collaboration for Theoretical Physics*	¥40M	2026-2029
FUTI Travel Award	¥350k	2026
JSPS KAKENHI 26K17136 (early-career), *Reconstruction of the Local Universe with Machine Learning*	¥3.5M	2026-2027
Royal Society International Collaboration Award Round 2 (Japan) 252140, with William Coulton, *Transformative joint analyses of transformative datasets*	£225k	2026-2029
Google Research Ecosystem Grant for IPMU CD3	¥11M	2025-2026
JSPS KAKENHI 24K22878 (research-activity startup), *AI staring into the void*	¥3M	2024-2025

TEACHING

Astro-AI Asia Network summer school, Taipei	organizer	2026
10th Vietnam School of Astrophysics, Quy Nhon	*Machine Learning*	2026
Lecture for UTokyo GUC	*Scientific Programming*	2026
Lectures for UTokyo class	*Cosmological Data Analysis*	2026
Astro-AI Asia Network summer school, Seoul	co-organizer	2025
Lecture for UTokyo GUC	*Scientific Programming*	2025
Lecture & Tutorial for ILANCE students, IPMU	*Introduction to Machine Learning*	2025
Astro-AI Asia Network summer school, Osaka	*Basic Deep Learning*	2024

SERVICE reviewer for ApJ, MNRAS, PRD, JCAP, NeurIPS ML4PS, ICML ML4Astro, ICML AI4Physics

ADVISING

Alexis Tastet (Ecole Polytechnique Masters)	visitor	2026
Miguel Ruiz Granda (UCantabria PhD)	assisted visitor	2026-
Kristers Nagainis (Latvia/Leiden PhD)	assisted visitor	2026-
Di He (UTokyo PhD)	assisted	2025-
Lillie Szemraj (Princeton undergrad)	assisted visitor	2025
Felicia Xiao (MIT undergrad)	visitor	2025
Baptiste Barthe-Gold (Ecole Polytechnique Masters)	visitor	2025
Jessica Cowell (Oxford DPhil → Manchester PD)	assisted	2024-2025
Akira Tokiwa (UTokyo PhD → industry)	assisted	2024-2025
Cooper Jacobus (UC Berkeley undergrad → Stanford PhD)	visitor	2024
Bonny Wang (CMU Masters → UChicago PhD)	visitor	2024-

PUBLICATIONS Citations: 1627, h-index: 21 (INSPIRE-HEP, as of 2026-10-04)

K.S. Croker, J. Cohon, V.T. Toth, M. Rinaldi, N. Fernandez, et al. (incl. **L. Thiele**), *A Strong Cosmologically Coupled Mass Growth Solution in Linearized General Relativity*, 2026,
arXiv:2609.38303 [gr-qc]

N. Aghanim, B. Maffei, J. Aumont, A. Beelen, B. Borgo, et al. (incl. **L. Thiele**), *FOSSIL: A future mission for CMB spectral distortion measurements*, 2026,
arXiv:2609.17618 [astro-ph.CO]

B. Cyr, N. Aghanim, E. Baker, E.S. Battistelli, R. Battye, et al. (incl. **L. Thiele**), *Peering Beyond the Veil of Last Scattering: A View of the Universe with CMB Spectral Distortions*, 2026,
arXiv:2609.16194 [astro-ph.CO]

Y. Itow, J. Liu, H. Maesaka, V. Mikuni, N.-M. Nguyen, et al. (incl. **L. Thiele**), *Future of Artificial Intelligence for Science in Japan 2024 Community Report*, 2026,
arXiv:2608.27807 [hep-ph]

J. Liu, V. Krishnaraj, K. Vovk, K. Aizawa, A.E. Bayer, et al. (incl. **L. Thiele**), *AI's Capability in Assisting Scientific Research in Physics, Astrophysics, and Cosmology II: Project Planning and Proposal Evaluation*, 2026,
arXiv:2607.25881 [cs.CL], oral at NeurIPS 2026 AI-Native Academia workshop

A. Hell, K. Vovk, V. Krishnaraj, J. Liu, K. Aizawa, et al. (incl. **L. Thiele**), *AI's Capability in Assisting Scientific Research in Physics, Astrophysics, and Cosmology I: Literature Review*, 2026,
arXiv:2607.25672 [astro-ph.IM], poster at NeurIPS 2026 AI-Native Academia workshop

B.Y. Wang, **L. Thiele**, M. Ho, *Cluster Mass Inference from Galaxy Kinematics*, 2026,
arXiv:2606.12938 [astro-ph.CO]

L. Thiele, *Machine Learning Techniques for Astrophysics and Cosmology: Simulation-Based Inference*, 2026,
arXiv:2605.10719 [astro-ph.CO], Invited chapter for the edited book "Machine Learning Techniques for Astrophysics and Cosmology" (Eds. C. Bambi, V. Kashyap, S. Shashank, N. Yoshida, Springer Singapore, expected in 2026)

A. Hell, **L. Thiele**, *LLMs with in-context learning for Algorithmic Theoretical Physics*, 2026,
arXiv:2605.08212 [cs.LG], poster at ICML 2026 AI4Physics workshop

L. Thiele, *Bayesian Cosmic Void Finding with Graph Flows*, 2026,
OJAp 9, arXiv:2602.14630 [astro-ph.CO], poster at ICML 2026 AI4Physics workshop

J. Armijo, **L. Thiele**, J. Liu, *Replicating weak-lensing summary-statistic covariances with normalizing flows*, 2026,
PRD 114, 023567, arXiv:2601.20669 [astro-ph.CO]

G. Fabbian, F. Bianchini, A. Sabyr, J.C. Hill, C.C. Lovell, **L. Thiele**, D.N. Spergel, *A new constraint on the y-distortion with FIRAS: implications for feedback models in galaxy formation and cosmic shear measurements*, 2025,
arXiv:2512.03038 [astro-ph.CO]

A. Tokiwa, A.E. Bayer, J. Armijo, J. Liu, R. Terasawa, **L. Thiele**, M. Alvarez, L. Blot, M. Takada, *Impact of Simulation Box Size for Weak Lensing: Replication and Super-Sample Effects*, 2025,
JCAP 07, 023 arXiv:2511.20423 [astro-ph.CO],

B. Barthe-Gold, N.-M. Nguyen, **L. Thiele**, *Reconstructing the local density field with combined convolutional and point cloud architecture*, 2025,
arXiv:2510.08573 [astro-ph.CO], poster at ML4PS workshop NeurIPS 2025

B.Y. Wang, **L. Thiele**, *Set-based Implicit Likelihood Inference of Galaxy Cluster Mass*, 2025,
arXiv:2507.20378 [cs.LG], spotlight talk at ML4Astro workshop (colocated with ICML 2025)

J.A. Cowell, J. Armijo, **L. Thiele**, G.A. Marques, C.P. Novaes, D. Grandón, S. Cheng, M. Shirasaki, D. Alonso, J. Liu, *First Constraints from Marked Angular Power Spectra with Subaru Hyper Suprime-Cam Survey First-Year Data*, 2025,
MNRAS 546, 2, arXiv:2507.12315 [astro-ph.CO]

L. Thiele, A.E. Bayer, N. Takeishi, *Simulation-Efficient Cosmological Inference with Multi-Fidelity SBI*, 2025,
arXiv:2507.00514 [astro-ph.CO], poster at ML4Astro workshop (colocated with ICML 2025)

E. Calabrese, J.C. Hill, H.T. Jense, A.L. Posta, I. Abril-Cabezas, et al. (incl. **L. Thiele**), *The Atacama Cosmology Telescope: DR6 Constraints on Extended Cosmological Models*, 2025,
JCAP 11, 063, arXiv:2503.14454 [astro-ph.CO]

C. Jacobus, **L. Thiele**, P. Harrington, J. Liu, Z. Lukic, *Enhancing Cosmological Simulations with Efficient and Interpretable Machine Learning in the Gabor Wavelet Basis*, 2024,
poster at Workshop on Machine Learning and the Physical Sciences (NeurIPS 2024)

L. Thiele, *De-baryonifying halos via optimal transport*, 2024,
arXiv:2411.18399 [astro-ph.CO]

J. Armijo, G.A. Marques, C.P. Novaes, **L. Thiele**, J.A. Cowell, D. Grandón, M. Shirasaki, J. Liu, *Cosmological constraints using Minkowski functionals from the first year data of the Hyper Suprime-Cam*, 2025,
MNRAS 537, 4, arXiv:2410.00401 [astro-ph.CO]

C.P. Novaes, **L. Thiele**, J. Armijo, S. Cheng, J.A. Cowell, G.A. Marques, E.G.M. Ferreira, M. Shirasaki, K. Osato, J. Liu, *Cosmology from HSC Y1 Weak Lensing with Combined Higher-Order Statistics and Simulation-based Inference*, 2025,
PRD 111, 8, arXiv:2409.01301 [astro-ph.CO]

A. Kogut, E. Switzer, D. Fixsen, N. Aghanim, J. Chluba, et al. (incl. **L. Thiele**), *The Primordial Inflation Explorer (PIXIE): Mission Design and Science Goals*, 2025,
JCAP 004, 020, arXiv:2405.20403 [astro-ph.CO]

S. Cheng, G.A. Marques, D. Grandón, **L. Thiele**, M. Shirasaki, B. Ménard, J. Liu, *Cosmological constraints from weak lensing scattering transform using HSC Y1 data*, 2025,
JCAP 01, 006, arXiv:2404.16085 [astro-ph.CO]

D. Grandón, G.A. Marques, **L. Thiele**, S. Cheng, M. Shirasaki, J. Liu, *Impact of baryonic feedback on HSC Y1 weak lensing non-Gaussian statistics*, 2024,
PRD 110, 103539, arXiv:2403.03807 [astro-ph.CO]

G.A. Marques, J. Liu, M. Shirasaki, **L. Thiele**, D. Grandón, K.M. Huffenberger, S. Cheng, J. Harnois-Déraps, K. Osato, W.R. Coulton, *Cosmology from weak lensing peaks and minima with Subaru Hyper Suprime-Cam survey first-year data*, 2023,
MNRAS 528, 3, arXiv:2308.10866 [astro-ph.CO]

L. Thiele, E. Massara, A. Pisani, C. Hahn, D.N. Spergel, S. Ho, B. Wandelt, *Neutrino mass constraint from an Implicit Likelihood Analysis of BOSS voids*, 2024,
ApJ 969, 89, arXiv:2307.07555 [astro-ph.CO]

L. Thiele, G.A. Marques, J. Liu, M. Shirasaki, *Cosmological constraints from HSC Y1 lensing convergence PDF*, 2023,
PRD 108, 123526, arXiv:2304.05928 [astro-ph.CO]

A.M. Delgado, D. Anglés-Alcázar, **L. Thiele**, M. Ntampaka, S. Pandey, K. Lehman, R.S. Somerville, S. Genel, F. Villaescusa-Navarro, *Predicting the impact of feedback on matter clustering with machine learning in CAMELS*, 2023,
MNRAS 526, 4, arXiv:2301.02231 [astro-ph.GA]

D. Wadekar, **L. Thiele**, J.C. Hill, S. Pandey, F. Villaescusa-Navarro, et al., *The SZ flux-mass (Y-M) relation at low halo masses: improvements with symbolic regression and strong constraints on baryonic feedback*, 2022,
MNRAS 522, 2, arXiv:2209.02075 [astro-ph.CO]

B.K.K. Lee, W. Coulton, **L. Thiele**, S. Ho, *An exploration of the properties of cluster profiles for the thermal and kinetic Sunyaev-Zel'dovich effects*, 2022,
MNRAS 517, 420, arXiv:2205.01710 [astro-ph.CO]

L. Thiele, M. Cranmer, W. Coulton, S. Ho, D.N. Spergel, *Predicting the Thermal Sunyaev-Zel'dovich Field using Modular and Equivariant Set-Based Neural Networks*, 2022,
MLST 3, 035002, `arXiv:2203.00026 [astro-ph.CO]`, poster at the Fourth Workshop on Machine Learning and the Physical Sciences (NeurIPS 2021)

L. Thiele, D. Wadekar, J.C. Hill, N. Battaglia, J. Chluba, F. Villaescusa-Navarro, L. Hernquist, M. Vogelsberger, D. Anglés-Alcázar, F. Marinacci, *Percent-level constraints on baryonic feedback with spectral distortion measurements*, 2022,
PRD 105, 083505, `arXiv:2201.01663 [astro-ph.CO]`

D. Wadekar, **L. Thiele**, F. Villaescusa-Navarro, J.C. Hill, D.N. Spergel, M. Cranmer, N. Battaglia, D. Anglés-Alcázar, L. Hernquist, S. Ho, *Augmenting astrophysical scaling relations with machine learning: application to reducing the SZ flux-mass scatter*, 2022,
PNAS 120(12), `arXiv:2201.01305 [astro-ph.CO]`

F. Villaescusa-Navarro, S. Genel, D. Anglés-Alcázar, L.A. Perez, P. Villanueva-Domingo, et al. (incl. **L. Thiele**), *The CAMELS project: public data release*, 2022,
ApJS 265, 54, `arXiv:2201.01300 [astro-ph.CO]`

B. Maffei, M.H. Abitbol, N. Aghanim, J. Aumont, E. Battistelli, et al. (incl. **L. Thiele**), *BISOU: a balloon project to measure the CMB spectral distortions*, 2021,
16th Marcel Grossmann Meeting, `arXiv:2111.00246 [astro-ph.IM]`

F. Villaescusa-Navarro, S. Genel, D. Anglés-Alcázar, **L. Thiele**, R. Dave, et al., *The CAMELS Multifield Dataset: Learning the Universe's Fundamental Parameters with Artificial Intelligence*, 2021,
ApJS 259, 61, `arXiv:2109.10915 [cs.LG]`

F. Villaescusa-Navarro, S. Genel, D. Anglés-Alcázar, D.N. Spergel, Y. Li, et al. (incl. **L. Thiele**), *Robust marginalization of baryonic effects for cosmological inference at the field level*, 2021,
`arXiv:2109.10360 [astro-ph.CO]`

F. Villaescusa-Navarro, D. Anglés-Alcázar, S. Genel, D.N. Spergel, Y. Li, et al. (incl. **L. Thiele**), *Multifield Cosmology with Artificial Intelligence*, 2021,
`arXiv:2109.09747 [astro-ph.CO]`

L. Thiele, Y. Guan, J.C. Hill, A. Kosowsky, D.N. Spergel, *Can small-scale baryon inhomogeneities resolve the Hubble tension? An investigation with ACT DR4*, 2021,
PRD 104, 063535, `arXiv:2105.03003 [astro-ph.CO]`

L. Thiele, J.C. Hill, K.M. Smith, *Accurate Analytic Model for the Weak Lensing Convergence One-Point Probability Distribution Function and its Auto-Covariance*, 2020,
PRD 102, 123545, `arXiv:2009.06547 [astro-ph.CO]`

L. Thiele, F. Villaescusa-Navarro, D.N. Spergel, D. Nelson, A. Pillepich, *Teaching neural networks to generate Fast Sunyaev Zel'dovich Maps*, 2020,
ApJ 902, 129, `arXiv:2007.07267 [astro-ph.CO]`

R. Cayuso, O.J.C. Dias, F. Gray, D. Kubizňák, A. Margalit, J.E. Santos, R.G. Souza, **L. Thiele**, *Massive vector fields in Kerr–Newman and Kerr–Sen black hole spacetimes*, 2020,
JHEP 159, `arXiv:1912.08224 [hep-th]`

L. Thiele, C.A.J. Duncan, D. Alonso, *Disentangling magnification in combined shear clustering analyses*, 2020,
MNRAS 491, 1746, `arXiv:1907.13205 [astro-ph.CO]`

R. Cayuso, F. Gray, D. Kubizňák, A. Margalit, R.G. Souza, **L. Thiele**, *Principal Tensor Strikes Again: Separability of Vector Equations with Torsion*, 2019,
PLB 795, 650, `arXiv:1906.10072 [hep-th]`

L. Thiele, J.C. Hill, K.M. Smith, *Accurate analytic model for the thermal Sunyaev-Zel'dovich one-point probability distribution function*, 2019,
PRD 99, 103511, `arXiv:1812.05584 [astro-ph.CO]`

F. Dinc, M. Medvidovic, **L. Thiele**, *Effective Geometry Monte Carlo: A Fast and Reliable Simulation Framework for Molecular Communication*, 2019,
IEEE Access 7, 28635

F. Dinc, **L. Thiele**, B.C. Akdeniz, *The effective geometry Monte Carlo algorithm: applications to molecular communication*, 2019,
PLA 383, 2594, `arXiv:1809.06438 [cs.ET]`

TALKS

Kavli Astrophysics Symposium, Cambridge		9/2026
Kavli Astrophysics Symposium, Cambridge	invited	9/2026
New Frontiers in Cosmology, A Coruna		8/2026
Renoir Science, CPPM Marseille	invited	5/2026
Sydney PPC Seminar, remote		4/2026
Fundamentals Seminar, IPMU		2/2026
AI4HEP East Asia, KEK Tsukuba	invited	1/2026
AI4KMI, Nagoya	invited	1/2026
3rd Shanghai Assembly		11/2025
28th IBIS, Naha	invited	11/2025
NPML conference, Tokyo	invited	10/2025
ML workshop @ McGill		9/2025
Mila, Montreal	invited	9/2025
21st Rencontres du Vietnam	invited	8/2025
AI for Fugaku seminar		7/2025
ML4Astro, UTokyo		5/2025
LeCosPA Meets IPMU, Taipei		4/2025
FOPM seminar, UTokyo		3/2025
Voids @ CPPM, Marseille	invited	2/2025
KICC, Cambridge UK		2/2025
FAIRS-Japan workshop Nagoya	invited	12/2024
11th KIAS workshop Gyeongju	invited	10/2024
Cosmo'24 Kyoto		10/2024
LSS Quest Osaka		6/2024
MPA cosmology seminar		5/2024
Yale cosmology seminar	invited	2/2024
AI4Phys @ IPMU		1/2024
CCA Cosmo x ML tristate meeting		10/2023
CMB constellation meeting KIPAC Stanford		10/2023
DESI lunch Berkeley Lab		10/2023
BCCP seminar UC Berkeley		10/2023
IPMU CD3 seminar		9/2023
Institute d'Astrophysique Spatiale Orsay		9/2023
Cosmo'23 Madrid		9/2023
Nagoya		4/2023

IPMU	3/2023
Princeton gravity group	2/2023
UCL Physics & Astronomy	9/2022
IAS astro coffee	3/2022
AAS 239	1/2022
Cosmology Talks	1/2022
Learn the Universe @ CCA	8/2021
CMB-S4 meeting	8/2021
MPA Garching	10/2020
German Astronomical Society	9/2020
Perimeter Institute	5/2020
Princeton/IAS cosmo lunch	5/2020
CCA Cosmo x ML	5/2020